Geometric Invariant Theory

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1 Group Schemes and Quotients

Let S be a fixed base scheme. All schemes, morphisms and fiber products are assumed to be over S.

Definition 1.1. A *group scheme* is a scheme G/S together with morphisms $\mu : G \times_S G \to G$ (multiplication), $\beta : G \to G$ (inversion), $e : S \to G$ (identity) such that

- *(associativity)*

$$\begin{tikzcd} G \times_S G \times_S G \arrow[r, "\mathrm{id}_G \times \mu"] \arrow[d, "\mu \times \mathrm{id}_G"'] & G \times_S G \arrow[d, "\mu"] \\ G \times_S G \arrow[r, "\mu"] & G \end{tikzcd}$$

  commutes,

- *(inverses)*

$$G \xrightarrow{\Delta} G \times_S G \underset{\beta \times \mathrm{id}_G}{\overset{\mathrm{id}_G \times \beta}{\rightrightarrows}} G \times_S G \xrightarrow{\mu} G$$

  both equal $e \circ (G \to S)$, where Δ is the diagonal.

- *(identity)* The compositions

$$\begin{tikzcd} & S \times_S G \arrow[dr, "e \times \mathrm{id}_G"] & & \\ G \arrow[ur, equal] \arrow[dr, equal] & & G \times_S G \arrow[r, "\mu"] & G \\ & G \times_S S \arrow[ur, "\mathrm{id}_G \times e"'] & & \end{tikzcd}$$

  both equal id_G.

Definition 1.2. An *algebraic group* G over a field k is a group scheme G/k which is an algebraic scheme, smooth over k.

Definition 1.3. An *action* of a group scheme G/S on a scheme X/S is a morphism $\sigma : G \times_S X \to X$ such that (i)

$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, "\mathrm{id}_G \times \sigma"] \arrow[d, "\mu \times \mathrm{id}_X"'] & G \times_S X \arrow[d, "\sigma"] \\ G \times_S X \arrow[r, "\sigma"] & X \end{tikzcd}$$

commutes, and (ii) the composition

$$X \simeq S \times_S X \xrightarrow{e \times \mathrm{id}_X} G \times_S X \xrightarrow{\sigma} X$$

equals id_X.

Definition 1.4. Let $f : T \to X$ be a T-valued point of X. Then define

$$\psi_f : G \times_S T \to X \times_S T$$

as $(\sigma \circ (\mathrm{id}_G \times f), \pi_T)$. In the special case of $f = \mathrm{id}_X$, we write $\Psi = \psi_{\mathrm{id}_X}$, which is simply $\Psi = (\sigma, \pi_X) : G \times_S X \to X \times_S X$.

The image of ψ_f, denoted $\mathrm{Orbit}(f)$, is called the *orbit* of f.

Using the canonical section $(f, \mathrm{id}_T) : T \to X \times_S T$, we can construct the fiber product $\mathrm{Stab}(f)$, which is called the *stabilizer* of f.

$$\begin{tikzcd}
\mathrm{Stab}(f) \arrow[r] \arrow[d] & T \arrow[d, "(f,\mathrm{id}_T)"] \\
G \times_S T \arrow[r, "\psi_f"] & X \times_S T
\end{tikzcd}$$

Definition 1.5. Given an action σ of G/S on X/S, a morphism $\pi : X \to Y$ is called a *categorical quotient* if

$$G \times_S X \underset{\pi_X}{\overset{\sigma}{\rightrightarrows}} X \xrightarrow{\pi} Y$$

is a coequalizer diagram. That is, the diagram commutes and for every $f : X \to Z$ with $f \circ \sigma = f \circ \pi_X$ there exists a unique $g : Y \to Z$ such that $f = g \circ \pi$. Note that a categorical quotient (if it exists) is unique up to unique isomorphism.

Definition 1.6. Given an action σ of G/S on X/S, a morphism $\pi : X \to Y$ is called a *geometric quotient* if

(*i*) $\pi \circ \sigma = \pi \circ \pi_X$,

(*ii*) π is surjective, and the image of Ψ is $X \times_Y X$ (or equivalently, the geometric fibers of π are precisely the orbits of the geometric points of X),

(*iii*) π is *submersive*, i.e. a subset $U \subset Y$ is open iff $\pi^{-1}(U) \subset X$ is open (i.e. Y has a quotient topology),

(*iv*) the structure sheaf $\mathcal{O}_Y$ is the subsheaf of $\pi_* \mathcal{O}_X$ consisting of invariant functions, i.e. if $\alpha \in \pi_* \mathcal{O}_X(U) = \mathcal{O}_X(\pi^{-1}(U))$ then $\alpha \in \mathcal{O}_Y(U)$ if and only if

$$\begin{tikzcd}
G \times \pi^{-1}(U) \arrow[r, "\sigma"] \arrow[d, "\pi_2"'] & \pi^{-1}(U) \arrow[d, "\alpha"] \\
\pi^{-1}(U) \arrow[r, "\alpha"] & \mathbb{A}^1
\end{tikzcd}$$

commutes.

Definition 1.7. Given an action σ of G on X, a morphism $\pi : X \to Y$ is called a *universal categorical quotient* (resp. *universal geometric quotient*) if for all morphisms $Y' \to Y$ the base change $\pi' : X' \to Y'$ is a categorical quotient (resp. geometric quotient). If this only holds for flat morphisms $Y' \to Y$, we call π a *uniform categorical quotient* (resp. *uniform geometric quotient*).

Remark 1.8. The problem with quotients, as I understand it, is that intuitively we would like to have a geometric quotient, but often only a categorical quotient exists. For example, take $X = \mathbb{A}^n$ and $G = \mathbb{G}_m$ (action by scaling). Then $\mathbb{A}^n \to \mathrm{pt}$ is a categorical quotient for sure, but not really what we expect.

If we take $X = \mathbb{A}^n - \{0\}$ and $G = \mathbb{G}_m$, then $\mathbb{A}^n - \{0\} \to \mathbb{P}^{n-1}$ is a geometric quotient, which is rather nice. In particular, we have the following proposition.

Proposition 1.9. *A geometric quotient $\pi : X \to Y$ is a categorical quotient. In particular, a universal geometric quotient is a universal categorical quotient.*

Proof. Let $f : X \to Z$ be an S-morphism with $f \circ \sigma = f \circ p_2$. Our goal is to construct a morphism $g : Y \to Z$ with $f = g \circ \pi$. Let $\{V_i\}$ be an affine open covering of Z. Then $f^{-1}(V_i)$ is an invariant open subset of X for each i, so by condition (ii) of Definition 1.6, $f^{-1}(V_i) = \pi^{-1}(U_i)$ for *some* subset $U_i \subset Y$. But then U_i is open by (iii), and since π is surjective, $\{U_i\}$ is an open cover of Y. Now note that any morphism $g : Y \to Z$ with $f = g \circ \pi$ must satisfy $g(U_i) \subset V_i$. Hence it must be defined by a set of (compatible) morphisms $h_i : \mathcal{O}_Z(V_i) \to \mathcal{O}_Y(U_i)$ such that

$$\begin{tikzcd}
\mathcal{O}_Z(V_i) \arrow[r, "h_i"] \arrow[d, "f^*"] & \mathcal{O}_Y(U_i) \arrow[d, "\pi^*"] \\
\mathcal{O}_X(f^{-1}(V_i)) \arrow[r, equal] & \mathcal{O}_X(\pi^{-1}(U_i))
\end{tikzcd}$$

commutes. Since π^* is injective by (iv), the h_i are unique if they exist. Well, for any $\alpha \in \mathcal{O}_Z(V_i)$ we have that $\pi^*(\alpha)$ is an invariant element of $\mathcal{O}_X(\pi^{-1}(U_i))$ in the sense of (iv), hence it lies in the subring $\pi^*(\mathcal{O}_Y(U_i))$, so such h_i exist. It remains to check that the h_i agree on overlaps, so that they indeed glue to the desired $g : Y \to Z$. $\square$

Lemma 1.10. *Let $\pi : X \to Y$ be a categorical quotient. Then, if X is reduced/connected/irreducible/locally integral/locally integral and normal, then so is Y, respectively.*

Proof. (Reduced) We have an inclusion $Y^{\mathrm{red}} \to Y$, and π lifts to $\pi^{\mathrm{red}} : X \to Y^{\mathrm{red}}$ as X is reduced. It is easy to see $\pi^{\mathrm{red}} \circ \sigma = \pi^{\mathrm{red}} \circ \pi_X$, so by the universal property we obtain a map $Y \to Y^{\mathrm{red}}$. Therefore, $Y = Y^{\mathrm{red}}$. (Connected) Let $Y_0 \subset Y$ be a connected component. Then must have $\pi^{-1} = X$, so can view $\pi : X \to Y_0$. The universal property now gives $Y \to Y_0$ and thus $Y = Y_0$. Et cetera. $\square$

Lemma 1.11. *The following are equivalent:*

(i) $\pi : X \to Y$ *is a universal categorical quotient,*

(ii) for all morphisms $Y' \to Y$ with Y' affine, the base change $\pi' : Y' \to X'$ is a categorical quotient,

(iii) there exists an affine open cover $\{U_i\}$ of Y such that each restriction $\pi : \pi^{-1}(U_i) \to U_i$ is a universal categorical quotient.

Lemma 1.12. *A morphism $\pi : X \to Y$ is a categorical quotient if the following conditions are satisfied:*

(i) $\pi \circ \sigma = \pi \circ \pi_X$,

(ii) $\mathcal{O}_Y$ *is the subsheaf of invariants of* $\pi_* \mathcal{O}_X$,

(iii) if $W \subset X$ *are invariant closed subsets, then* $\pi(W)$ *is closed in* Y*, and for any such* W_i *we have*

$$\pi\left(\bigcap_i W_i\right) = \bigcap_i \pi(W_i).$$

Moreover, in this case π *is submersive.*

Proof. (Write proof) □

Definition 1.13. Let G be a group acting via σ on a scheme X. The action σ is said to be

- *closed* if for all geometric points x of X, the orbit $\mathrm{Orbit}(x) \subset \overline{X}$ is closed,
- *separated* if the image of $\Psi = (\sigma, \pi_X) : G \times_S X \to X \times_S X$ is closed,
- *proper* if Ψ is proper,
- *free* if Ψ is a closed immersion.

Lemma 1.14. *If a geometric quotient* $\pi : X \to Y$ *exists, then the action is closed. In this case* Y/S *is separated if and only if* σ *is separated.*

Lemma 1.15. *Let* $\pi : X \to Y$ *be a geometric quotient. If* $S = \operatorname{Spec} k$ *and* G, X, Y *are algebraic* k*-schemes, and* G *is affine and* σ *is proper, then* π *is affine.*

Definition 1.16. Let $\pi : X \to Y$ be a geometric quotient. Assume that G is flat and of finite type over S. Then X is a *principal fiber bundle* over Y, with group G, if

(i) π is a flat morphism of finite type,

(ii) Ψ is an isomorphism $G \times_S X \xrightarrow{\sim} X \times_Y X$.

Proposition 1.17. *Let* $S = \operatorname{Spec} k$*,* G *an algebraic group,* X, Y *algebraic schemes, and* $\pi : X \to Y$ *a geometric quotient. Then* X *is a principal fiber bundle over* Y *with group* G*.*

2 Reductive Groups

Let X be a scheme over a fixed field k (not necessarily algebraically closed), with action of G. Our purpose is to investigate for which open invariant subsets $U \subset X$ a geometric quotient U/G exists.

Definition 2.1. Let G be an algebraic group. A *representation* is a morphism $\rho : G \to \mathrm{GL}_n(k)$ that respects multiplication. It can also be viewed as a $k[G]$-module, where G acts via ρ.

Such a representation induces *three* types of actions. First, an action on affine n-space: $G \times \mathbb{A}^n \to \mathbb{A}^n$. Secondly, on vector spaces $G(k) \times V \to V$, where V is an n-dimensional k-vector space. Thirdly:

Definition 2.2. Let G be a linear algebraic group. Write $S = \mathcal{O}_G(G)$ and $\alpha : S \to S \otimes S$ for the multiplication map, and $\beta : S \to k$ the inclusion of the identity element. Let V be a vector space (or ring) over k. Then a *dual action* of G on V is a morphism of vector spaces (or rings)

$$\hat{\sigma} : V \to S \otimes_k V$$

such that

$$\begin{tikzcd}
 & S \otimes_k V \arrow[dr, "\alpha \otimes \mathrm{id}_V"] & \\
V \arrow[ur, "\hat{\sigma}"] \arrow[dr, "\hat{\sigma}"'] & & S \otimes_k S \otimes_k V \\
 & S \otimes_k V \arrow[ur, "\mathrm{id}_V \otimes \alpha"'] &
\end{tikzcd}$$

commutes and

$$V \xrightarrow{\hat{\sigma}} S \otimes_k V \xrightarrow{\beta \otimes \mathrm{id}_V} V$$

equals id_V.

Remark 2.3. When V is a vector space, this notion of a dual action is just a differently worded definition of a representation. Similarly, when $V = R$ is a ring, this dual action is just an action of G on the scheme $\operatorname{Spec} R$.

Definition 2.4. Let $\hat{\sigma}$ be a dual action of G on V. Then a vector space $W \subset V$ is *invariant* under the action of G if $\hat{\sigma}(W) \subset S \otimes_k W$. Moreover, $v \in V$ is *invariant* if $\hat{\sigma}(x) = 1 \otimes x$.

Definition 2.5. An algebraic group G is *reductive* if its radical is a torus, and *linearly reductive* if every representation of G is semisimple (i.e. completely reducible).

Remark 2.6. Sometimes being linear reductive is defined as the exactness of the functor $\bullet^G : \mathrm{QCoh}([\mathrm{pt}/G]) \to \mathrm{QCoh}(\mathrm{pt}), V \mapsto V^G$. Indeed, if $0 \to V \to W \to U \to 0$ is an exact sequence of $k[G]$-modules, semisimplicity of W yields a splitting, and exactness of $0 \to V^G \to W^G \to U^G \to 0$ easily follows. Conversely, if G is not linearly reductive, then there exists some non-semisimple $k[G]$-module W, with submodule $V \subset W$ such that $W \not\simeq V \oplus (W/V)$. So we have the exact sequence of $k[G]$-modules $0 \to V \to W \to (W/V) \to 0$, but its image under $\bullet^G$ will not be exact.

Example 2.7. The groups $\mathbb{G}_m, \mathrm{GL}_n(\mathbb{C}), \mathrm{SL}_n(\mathbb{C})$ and $\mathrm{PGL}_n(\mathbb{C})$ are all reductive. Also any finite group is reductive. A group which is not reductive is $\mathbb{G}_a$: since it is solvable (it is abelian) its radical is $\mathbb{G}_a$, which is not a torus. Also, we see that it is not linearly reductive: take $V = k^2$ with $\mathbb{G}_a$ acting via $t \cdot (x, y) = (x + y, y)$. Then $U := k \cdot (1, 0) \subset V$ is a sub-$k[\mathbb{G}_a]$-module, but $V \not\simeq U \oplus (V/U)$.

Definition 2.8. Let G be a linearly reductive group, $S = \mathcal{O}_G(G)$, and $\hat{\sigma}$ a dual action of G on a vector space V. Then a *Reynolds operator* is a morphism $E : V \to V$ such that

- E commutes with $\hat{\sigma}$, i.e. $\hat{\sigma} \circ E = (\mathrm{id}_S \otimes E) \circ \hat{\sigma}$,
- $E^2 = E$,
- $Ev = v$ if and only if v is invariant.

In the case that V is finite-dimensional, this E is the projection onto the subspace of V where G acts trivially.

The same holds when $V = R$ is a ring. The image of E is the subspace of invariant elements of R, which is a subring R_0. Although E is not a ring morphism, it does satisfy

$$\text{If } x \in R, \quad y \in R_0, \quad \text{then } E(xy) = (Ex)y.$$

3 The Affine GIT Quotient

In case X is affine and G is reductive, regarding quotients of X by G, this theorem is really all that you need.

Theorem 3.1 (The affine case)**.** *Let X be an affine scheme over k, let G be a reductive algebraic group, and let σ be an action of G on X. Then a uniform categorical quotient $\pi : X \to Y$ exists, π is universally submersive, and Y is affine. Moreover, if X is algebraic, then Y is algebraic over k. If k is of characteristic zero, then this is a universal categorical quotient. Moreover, if X is noetherian, then Y is noetherian.*

Proof. This is using the affine GIT quotient $Y := \operatorname{Spec}(R) /\!\!/ G = \operatorname{Spec}\left(R^G\right)$ □

Example 3.2. For finite groups, this construction seems to work fine. E.g. let $X = \mathbb{A}^1$ and $G = \mathbb{Z}/2\mathbb{Z}$ acting by $x \mapsto -x$. Then the affine GIT quotient is

$$X /\!\!/ G = \operatorname{Spec}(k[x]^{\mathbb{Z}/2\mathbb{Z}}) = \operatorname{Spec}(k[x^2]) = \mathbb{A}^1,$$

and the quotient map is given by

$$X \to X /\!\!/ G, \quad x \mapsto x^2.$$

Example 3.3. Take $X = \mathbb{A}^3$ and $G = S_3$ acting by permuting the coordinates. Then the affine GIT quotient is

$$X /\!\!/ G = \operatorname{Spec}(k[x,y,z]^{S_3}) = \operatorname{Spec}(k[x+y+z, x^2+y^2+z^2, xyz]) = \operatorname{Spec}(k[u,v,w]/(?)).$$

Remark 3.4. Still this is not quite the analysis that we want to have. At this point, the best we can do for $X = \mathbb{A}^n$ and $G = \mathbb{G}_m$ is still the (categorical) quotient $\mathbb{A}^n \to \operatorname{Spec}(k[x_1, \ldots, x_n]^{\mathbb{G}_m}) = \operatorname{Spec}(k) = \mathrm{pt}$.

Rather what we want to have, is a quotient like $X = \mathbb{A}^n - \{0\}$ by $G = \mathbb{G}_m$,

$$\mathbb{A}^n - \{0\} \to \mathbb{P}^{n-1}.$$

4 The Projective GIT Quotient

An action of a reductive group G on a projective variety $X \subset \mathbb{P}^n$ is said to be *linear* if G acts via a homomorphism $G \to \mathrm{GL}_{n+1}$.

Suppose we have some reductive group G acting linearly on a projective variety $X \subset \mathbb{P}^n$. We let $X /\!\!/ G$ denote the projective variety $\mathrm{Proj}(R^G)$ associated to the finitely generated graded k-algebra R^G of G-invariant functions where $R = R(X)$ is the homogeneous coordinate ring of X. It is important to remark that this is dependent of the choice of embedding $X \subset \mathbb{P}^n$. (In fact, later we will see this quotient depends on the choice of an invertible sheaf $\mathcal{L}$ and *linearization* of this sheaf.) The inclusion $R^G \to R$ defines a rational map

$$\varphi : X \dashrightarrow X /\!\!/ G$$

which is undefined only on the null cone

$$X^{\mathrm{us}} := \{x \in X : f(x) = 0 \text{ for all } f \in R^G_+\}.$$

We call these points *unstable* points, and points in the complement $X^{\mathrm{ss}} := X - X^{\mathrm{us}}$ are called the *semi-stable* points.

The *projective GIT quotient* for the linear action of G on $X \subset \mathbb{P}^n$ is the morphism $\varphi : X^{\mathrm{ss}} \to X /\!\!/ G$.

Note that for any $f \in R^G_+$ we have that

$$A((X /\!\!/ G)_f) = \left(\left(R^G\right)_f\right)_0 = \left((R_f)_0\right)^G = A(X_f)^G,$$

from which we see that on the affine patches, the projective GIT quotient restricts to the affine GIT quotient. Therefore, we can conclude that is a good quotient and a categorical quotient.

We can now ask if there is an open subset $X^{\mathrm{s}} \subset X^{\mathrm{ss}}$ on which the quotient becomes a geometric quotient. This is the case for

$$X^{\mathrm{s}} := \left\{ \begin{array}{c} x \in X : \quad \dim G_x = 0 \text{ and there exists } f \in R(X)^G_+ \text{ such that} \\ x \in X_f \text{ and the action of } G \text{ on } X_f \text{ is closed} \end{array} \right\}$$

Example 4.1. Consider the linear action of $\mathbb{G}_m$ on $X = \mathbb{P}^n$ given by

$$t \cdot (x_0 : x_1 : \cdots : x_n) = (t^{-1}x_0 : tx_1 : \cdots : tx_n).$$

In this case $R(X) = k[x_0, \ldots, x_n]$, and the G-invariant functions are

$$R(X)^G = k[x_0x_1, x_0x_2, \ldots, x_0x_n] \simeq k[y_0, \ldots, y_{n-1}],$$

which corresponds to the projective variety $X /\!\!/ G = \mathbb{P}^{n-1}$. We obtain the rational map

$$\begin{array}{ccc} \varphi : X = \mathbb{P}^n & \dashrightarrow & X /\!\!/ G = \mathbb{P}^{n-1} \\ (x_0 : x_1 : \cdots : x_n) & \longmapsto & (x_0x_1 : x_0x_2 : \cdots x_0x_n) \end{array}$$

and we see that the null cone is given by

$$X^{\mathrm{us}} = \{(x_0 : \cdots : x_n) : x_0 = 0 \text{ or } x_1 = x_2 = \cdots = x_n = 0\}.$$

In particular, its complement is

$$X^{\mathrm{ss}} = \{(x_0 : \cdots : x_n) : x_0 \neq 0 \text{ and } (x_1, x_2, \ldots, x_n) \neq 0\} \simeq \mathbb{A}^n - \{0\}.$$

Hence

$$\varphi : \mathbb{A}^n - \{0\} \to \mathbb{P}^{n-1}$$

is a good quotient for the action of $\mathbb{G}_m$ on X^{ss}. As the preimage of each point in $X /\!\!/ G$ is a single orbit, this is also a geometric quotient. Moreover, every point is stable as all orbits are closed in $\mathbb{A}^n - \{0\}$ and have zero-dimensional stabilizers.

However, in general it is hard to determine which points are the (semi)-stable points, as finding generators for the invariant algebra is hard. For this we need other criteria.

5 A Topological Criterion

Let G be a reductive group acting linearly on a projective variety $X \subset \mathbb{P}^n$. This induces an action of G on the affine cone $\tilde{X} \subset \mathbb{A}^{n+1}$.

Proposition 5.1. *Let $\tilde{x} \in \tilde{X}$ be a point lying over $x \in X$. Then*

(i) $x \in X$ is semi-stable if and only if $0 \notin \overline{G \cdot \tilde{x}}$,

(ii) $x \in X$ is stable if and only if $\dim G_{\tilde{x}} = 0$ and $G \cdot \tilde{x}$ is closed in $\tilde{X}$.

Proof. (i) By definition, if x is semi-stable, there exists a homogeneous G-invariant polynomial $f \in R(X)^G$ with $f(x) \neq 0$. Viewing f as a G-invariant function on $\tilde{X}$, we have $f(\tilde{x}) \neq 0$ and thus f must be constantly non-zero on the whole $\overline{G \cdot \tilde{x}}$, in particular $0 \notin \overline{G \cdot \tilde{x}}$. Conversely, if $\overline{G \cdot \tilde{x}}$ and 0 are disjoint G-invariant closed subsets, there must exist a G-invariant polynomial $f \in R(\tilde{X})^G$ such that $f(\overline{G \cdot \tilde{x}}) = 1$ and $f(0) = 0$ (this requires an argument though). Now just take some homogeneous piece of f which is non-zero on $\overline{G \cdot \tilde{x}}$ to show x is semi-stable.

(ii) (Write proof) □

Example 5.2. Let $X = \mathbb{P}^1$ and consider the action of $G = \mathbb{G}_m$ given by $t \cdot (x : y) = (tx : t^{-1}y)$. Applying the above criterion we have

$$(x : y) \text{ is semi-stable} \iff 0 \notin \overline{\mathbb{G}_m \cdot (\tilde{x}, \tilde{y})}$$

and after some thinking we conclude $X^{\mathrm{ss}} = \mathbb{P}^1 - \{(1 : 0), (0 : 1)\}$. Furthermore,

$$(x : y) \text{ is stable} \iff \dim G_{(\tilde{x},\tilde{y})} = 0 \text{ and } G \cdot (\tilde{x}, \tilde{y}) \text{ is closed}$$

and after some more thinking we conclude $X^{\mathrm{s}} = \mathbb{P}^1 - \{(1 : 0), (0 : 1)\}$ as well. Hence, we obtain the GIT quotient $\varphi : X^{\mathrm{ss}} = X^{\mathrm{s}} \to \star$, which is a geometric quotient. (This is a special case of the previous example)

As remarked before, the above construction depends on the embedding $X \subset \mathbb{P}^n$, and an abstract projective variety does not come with such a specified embedding. However, an ample line bundle $\mathcal{L}$ on X (or more precisely some power of $\mathcal{L}$) determines an embedding of X into a projective space (e.g. the $\mathcal{O}(1)$-sheaf). In order to construct a GIT quotient of an abstract projective variety X we need the extra data of a lift of the G-action to a line bundle on X, such a choice is called a *linearization* of the action.

6 Linearization of an Invertible Sheaf

Definition 6.1. Let $G, X, \mathcal{L}, \sigma$ be an algebraic group, an algebraic scheme, an invertible sheaf on X and an action of G on X. Then a *G-linearization* of $\mathcal{L}$ consists of an isomorphism

$$\phi : \sigma^*\mathcal{L} \xrightarrow{\sim} \pi_X^*\mathcal{L}$$

of sheaves on $G \times X$, satisfying the *cocycle condition*:

$$\pi_{23}^*\phi \circ (\mathrm{id}_G \times \sigma)^*\phi = (\mu \times \mathrm{id}_X)^*\phi.$$

Remark 6.2. This definition may seem abstract, because it is. It becomes more intuitive when viewing the invertible sheaf $\mathcal{L}$ as a line bundle L. Namely then, a linearization of the action of G w.r.t. L is an action of G on L such that

(i) for all $g \in G$ and $\ell \in L$, we have $\pi(g \cdot \ell) = g \cdot \pi(\ell)$,

(ii) for all $x \in X$ and $g \in G$ the map of fibers $L_x \to L_{g \cdot x}$ is a linear map.

There is another action induced by a G-linearization of $\mathcal{L}$, which is very important: a dual action of G on $H^0(X, \mathcal{L})$, which is given by the composition

$$H^0(X, \mathcal{L}) \xrightarrow{\sigma^*} H^0(G \times X, \sigma^*\mathcal{L}) \xrightarrow{\phi} H^0(G \times X, \pi_X^*\mathcal{L}) \simeq H^0(G, \mathcal{O}_G) \otimes H^0(X, L) \ .$$

This allows us to speak of *invariant sections* of $\mathcal{L}$.

Example 6.3. The easiest example would be $G = \mathrm{GL}_{n+1}$ (or $G = \mathrm{SL}_{n+1}$) acting naturally on $X = \mathbb{P}^n$, with $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(1)$. Note that $\mathcal{L}$ corresponds to the line bundle L which is the blowup of $\mathbb{A}^{n+1}$ at the origin, with the obvious projection $\pi : L \to \mathbb{P}^n$. Now the natural action

$$\mathrm{GL}_{n+1} \times L \to L$$

is easily seen to be a linearization. Also, it can be seen as

$$\begin{aligned} \phi : \sigma^*\mathcal{O}_{\mathbb{P}^n}(1) &\xrightarrow{\sim} p_2^*\mathcal{O}_{\mathbb{P}^n}(1) \\ \sigma^*(X_i) &\mapsto \sum_{j=0}^{n} a_{ij} p_2^*(X_i), \end{aligned}$$

where the a_{ij} are coefficients of the matrices of GL_{n+1}, seen as regular functions on $G \times \mathbb{P}^n$ (you can just pull them back from the regular functions on G). Note that this is just a very complicated way of denoting matrix multiplication.

Example 6.4. Let X be a variety, and consider the trivial line bundle $L = X \times k$. Then a linearization of a G-action on X w.r.t. L corresponds to a character $\chi : G \to \mathbb{G}_m$. The character χ defines a lift of the action to L given by

$$\begin{aligned} G \times L &\to L \\ (g, (x, z)) &\mapsto (g \cdot x, \chi(g)z). \end{aligned}$$

More generally, we can use a character $\chi : G \to \mathbb{G}_m$ to modify any given linearization L of G-action on X by defining

$$g \cdot (x, z) = (g \cdot x, \chi(g)(g \cdot z))$$

for $x \in X$ and $z \in L_x$.

Example 6.5. Forgetting the G-linearization yields a homomorphism

$$\nu : \mathrm{Pic}^G(X) \to \mathrm{Pic}(X),$$

whose kernel $\ker \nu$ consists of the G-linearization of the trivial line bundle $\mathcal{O}_X$. Such a linearization is given by a morphism $c : G \times X \to \mathbb{G}_m$ such that

$$c(gh, x) = c(g, hx)c(h, x)$$

for all $g, h \in G$ and $x \in X$. If we define $\gamma : G \to \mathcal{O}_X^*(X)$ by $\gamma(g)(x) = c(g^{-1}, x)$, this condition translates into the usual cocycle condition

$$\gamma(gh) = {}^g\gamma(h)\gamma(g),$$

where $({}^g u)(x) = u(g^{-1}x)$. Any change of trivialization is given by multiplication by some $u \in \mathcal{O}_X^*(X)$, which transforms $\gamma(g)$ into $\gamma'(g) = \gamma(g)^g u u^{-1}$, so it follows that

$$\ker \nu \simeq H^1(G, \mathcal{O}_X(X)^*),$$

and we obtain an exact sequence

$$1 \to H^1(G, \mathcal{O}_X(X)^*) \to \mathrm{Pic}^G(X) \xrightarrow{\nu} \mathrm{Pic}(X).$$

Example 6.6. Show that $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(1)$ does not admit a PGL_{n+1}-linearization! See page 32-33 of Mumford.

The idea is that the action

$$\sigma : \mathrm{PGL}_{n+1} \times \mathbb{P}^n \to \mathbb{P}^n$$

can be defined by the condition that

$$\sigma^* \mathcal{O}_{\mathbb{P}^n}(1) \simeq p_1^* \mathcal{O}_{\mathbb{P}^{n^2+2n}}(1) \otimes p_2^* \mathcal{O}_{\mathbb{P}^n}(1)$$
$$\sigma^*(X_i) \mapsto \sum_{j=0}^{n} p_1^*(a_{ij}) \otimes p_2^*(X_j).$$

But now, it is impossible to construct an isomorphism of sheaves $\sigma^*\mathcal{L} \to p_2^*\mathcal{L}$, since $p_1^*\mathcal{O}_{\mathbb{P}^{n^2+2n}}(1)$ has non-trivial class in the Picard group.

Note that the tensor product and inverse of G-linearized invertible sheaves carry natural G-linearizations themselves. Therefore, the set of G-linearized invertible sheaves modulo isomorphism, forms an abelian group, denoted

$$\mathrm{Pic}^G(X).$$

Moreover, if $f : X \to Y$ is a G-linear morphism of schemes on which G acts, there is an induced homomorphism

$$f^* : \mathrm{Pic}^G(X) \to \mathrm{Pic}^G(Y).$$

7 GIT using Linearized Invertible Sheaves

Let X be a scheme with action of a reductive algebraic group G, denoted by σ.

Definition 7.1. Let x be a geometric point of X. Then x is *pre-stable* (w.r.t. σ) if there exists an affine open neighborhood $U \subset X$ of x which is closed under the action of G.

Now let $\mathcal{L}$ be an invertible sheaf with G-linearization ϕ. Then x is *semi-stable* (w.r.t. $\sigma, \mathcal{L}, \phi$) if there exists a section $s \in H^0(X, \mathcal{L}^n)$ for some n such that $s(x) \neq 0$, X_s is affine, and s is invariant, i.e. if $\phi_n : \sigma^*(\mathcal{L}^n) \to p_2^*(\mathcal{L}^n)$ is induced by ϕ, then $\phi_n(\sigma^*(s)) = p_2^*(s)$.

Finally, x is *stable* (w.r.t. $\sigma, \mathcal{L}, \phi$) if there exists a section $s \in H^0(X, \mathcal{L}^n)$ for some n, such that $s(x) \neq 0$, X_s is affine, s is invariant and the action of G on X_s is closed. (Stability is also defined as having finite stabilizer)

The sets of pre-stable/semi-stable/stable geometric points are the sets of geometric points of an open subset of X. The corresponding 3 open subsets will be written

$$X^{\mathrm{s}}(\mathrm{Pre}), \qquad X^{\mathrm{ss}}(\mathcal{L}), \qquad X^{\mathrm{s}}(\mathcal{L}),$$

respectively.

Theorem 7.2. *Let X be an algebraic scheme over k, and let G be a reductive algebraic group acting on X. Suppose $\mathcal{L}$ is a G-linearized invertible sheaf on X. Then a uniform categorical quotient $\pi : X^{ss}(\mathcal{L}) \to Y$ exists. Moreover,*

- *π is affine and universally submersive,*
- *there is an ample invertible sheaf $\mathcal{M}$ on Y such that $\pi^*\mathcal{M} \simeq \mathcal{L}^n$ for some n (hence Y is a quasi-projective algebraic scheme),*
- *there is an open subset $\tilde{Y} \subset Y$ such that $X^s(\mathcal{L}) = \pi^{-1}(\tilde{Y})$ and such that $\pi|_{X^s(\mathcal{L})} : X^s(\mathcal{L}) \to \tilde{Y}$ is a uniform geometric quotient.*

Proof. One can construct the projective GIT quotient

$$Y := X /\!\!/_{\mathcal{L}} G = \operatorname{Proj}\left(\bigoplus_{n \geq 0} H^0(X, \mathcal{L}^n)^G\right).$$

□

Example 7.3. Note that the affine GIT quotient can be seen as a particular case of the projective GIT quotient (the trivial one). Given $X = \operatorname{Spec} R$ with action of G, one can take $\mathcal{L} = \mathcal{O}_X$ and as linearization $\phi : \sigma^*\mathcal{O}_X \to p_2^*\mathcal{O}_X$ the identity morphism. Then

$$H^0(X, \mathcal{O}_X^n)^G = R^G \qquad \text{for any } n \geq 0,$$

and we have

$$X /\!\!/_{\mathcal{L}} G = \operatorname{Proj}\left(\bigoplus_{n\geq 0} R^G\right) = \operatorname{Proj}\left(R^G[x]\right) = \mathbb{P}^0_{\operatorname{Spec}(R^G)} = \operatorname{Spec}\left(R^G\right),$$

which is the affine GIT quotient. Now in this case we then have $X^{\mathrm{ss}} = X$ as $1 \in R$ is an invariant section which does not vanish on any point.

Remark 7.4. In general, we have $H^0(X, \mathcal{L}^0)^G = \mathcal{O}_X(X)^G$ and hence there is always an induced map

$$X /\!\!/_{\mathcal{L}} G \to \operatorname{Spec}\left(\mathcal{O}_X(X)^G\right)$$

to the affine GIT quotient.

Remark 7.5. Note that $X^{\mathrm{ss}}(\mathcal{L})$ equals $X^{\mathrm{ss}}(\mathcal{L}^k)$ for any $k \geq 1$, and similarly for X^{s}. (Explain how their GIT quotients are related. I think they are the same, but am not sure how to proof this.)

8 Stability Analysis

Definition 8.1. A *one parameter subgroup* (1-PS) of G is a non-trivial homomorphism $\lambda : \mathbb{G}_m \to G$.

Let X be an algebraic scheme, *proper* over k, with action σ of G. Let x be a closed point of X, and let λ be a 1-PS of G.

Consider the morphism $\psi_x \circ \lambda$ from $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ to X, sending t to $\lambda(t) \cdot x$. Viewing $\mathbb{G}_m \subset \mathbb{A}^1$, this extends uniquely to a morphism $f : \mathbb{A}^1 \to X$ using the valuative criterion of properness of X over k. The closed point $f(0)$ is called the *specialization* of x as $t \to 0$, and is also denoted

$$\lim_{t\to 0} \lambda(t) \cdot x.$$

Clearly $f(0)$ is fixed under the action of $\mathbb{G}_m$.

Now take $\mathcal{L} \in \operatorname{Pic}^G(X)$, then we can consider the induced $\mathbb{G}_m$-linearized invertible sheaf on $f(0)$, which is given by a *character* of $\mathbb{G}_m$:

$$\chi : \mathbb{G}_m \to \mathbb{G}_m = \operatorname{Aut}(\mathcal{L}|_{f(0)}) : \alpha \mapsto \alpha^r, \qquad r \in \mathbb{Z}.$$

Definition 8.2. We define the *weight* of λ at x (w.r.t. $\mathcal{L}$) as

$$\mu^{\mathcal{L}}(x, \lambda) = -r.$$

(Note that the definition changed sign through the years)

Some immediate properties are:

- $\mu^{\mathcal{L}}(g \cdot x, \lambda) = \mu^{\mathcal{L}}(x, g^{-1}\lambda g)$ for any $g \in G$,
- for fixed x and λ, this defines a group morphism

$$\mu^{\mathcal{L}}(x, \lambda) : \operatorname{Pic}^G(X) \to \mathbb{Z}.$$

- if $f : X \to Y$ is a G-linear morphism of schemes, $\mathcal{L} \in \mathrm{Pic}^G(Y)$ and x a closed point of X, then

$$\mu^{f^*\mathcal{L}}(x, \lambda) = \mu^{\mathcal{L}}(f(x), \lambda).$$

- if $y = \lim_{t \to 0} \lambda(t) \cdot x$, then $\mu^{\mathcal{L}}(x, \lambda) = \mu^{\mathcal{L}}(y, \lambda)$.

The goal is to prove this theorem:

Theorem 8.3 (Hilbert–Mumford Numerical Criterion)**.** *Let G be a reductive group acting on X proper over k. Let $\mathcal{L} \in \mathrm{Pic}^G(X)$ with $\mathcal{L}$ ample. Then for any closed point x of X:*

$$x \in X^{\mathrm{ss}}(\mathcal{L}) \iff \mu^{\mathcal{L}}(x, \lambda) \geq 0 \text{ for all 1-PS } \lambda,$$
$$x \in X^{\mathrm{s}}_{(0)}(\mathcal{L}) \iff \mu^{\mathcal{L}}(x, \lambda) > 0 \text{ for all 1-PS } \lambda.$$

(Note that the subscript (0) indicates the stabilizers of the points are finite. Some authors use this as the definition of a stable point.)

Example 8.4. Consider the case $X = \mathbb{P}^n$ (or $X \subset \mathbb{P}^n$ might work as well) and G acting linearly on X. We mean that $\mathcal{O}_{\mathbb{P}^n}(1)$ is G-linearized.

Note that as before, there is a dual action

$$\hat{\sigma} : H^0(X, \mathcal{O}(1)) \to \mathcal{O}_G(G) \otimes H^0(X, \mathcal{O}(1))$$
$$X_i \mapsto \sum_{j=0}^{n} a_{ij} \otimes X_j$$

given by some $a_{ij} \in \mathcal{O}_G(G)$. This now induces a linear action on $\mathbb{A}^{n+1}$

$$\tilde{\sigma} : G \times \mathbb{A}^{n+1} \to \mathbb{A}^{n+1}.$$

Note that this action is compatible with σ, in the sense that

$$\begin{tikzcd} G \times \left(\mathbb{A}^{n+1} - \{0\}\right) \arrow[r, "\tilde{\sigma}"] \arrow[d] & \mathbb{A}^{n+1} - \{0\} \arrow[d] \\ G \times \mathbb{P}^n \arrow[r, "\sigma"] & \mathbb{P}^n \end{tikzcd}$$

commutes.

Moreover, for any 1-PS λ, choose a basis of $\mathbb{A}^{n+1}$ such that λ is diagonal (yes, this is possible), with $\lambda(t) = \mathrm{diag}(t^{r_0}, \ldots, t^{r_n})$. Then

$$\mu^{\mathcal{O}(1)}(x, \lambda) = \max\{-r_i : \tilde{x}_i \neq 0\}.$$

Even more, $\lim_{t \to 0} \lambda(t) \cdot \tilde{x}$ does not exist (resp. does exist, resp. is zero) if and only if $\mu(x, \lambda) > 0$ (resp. $\mu(x, \lambda) = 0$, resp. $\mu(x, \lambda) < 0$). Also, from this we immediately the nice interpretation of μ,

$$\mu^{\mathcal{O}(1)}(x, \lambda) = \min\{\alpha \in \mathbb{Z} : \lim_{t \to 0} t^{\alpha} \lambda(t) \cdot \tilde{x} \text{ exists}\}.$$

Example 8.5. Let Γ be a finitely generated group, $G \subset \mathrm{GL}_n(k)$ a reductive algebraic group, and let $X = \mathrm{Hom}(\Gamma, G)$ be the G-representation variety. It naturally has the structure of an algebraic variety by considering it as a closed subvariety of G^m, where m is the number of generators of Γ. Now G acts on X by conjugation, $g \cdot \rho = g\rho g^{-1}$.

Let $\lambda : \mathbb{G}_m \to G$ be any 1-PS. After choosing a suitable basis, we can assume that

$$\lambda(t) = \mathrm{diag}\,(t^{r_1}, \ldots, t^{r_n}) \qquad \text{with } r_1 \geq \cdots \geq r_n.$$

Now we see that the action of λ on any $\rho = (A_1, \ldots, A_m)$ is given by

$$\lambda(t) \cdot \rho = \lambda(t)\rho\lambda(t)^{-1} = (A'_1, \ldots, A'_m) \quad \text{with} \quad (A'_\ell)_{ij} = (A_\ell)_{ij} \cdot t^{r_i - r_j}.$$

In particular, since any A_ℓ has at least some non-zero entry either on or under the diagonal, where $i \geq j$ so $r_i - r_j \leq 0$, it follows that $\mu(\rho, \lambda) = \min\{\alpha \in \mathbb{Z} : \lim_{t\to 0} t^\alpha \lambda(t) \cdot \rho \text{ exists }\} \geq 0$. This shows that all points $\rho \in X$ are semi-stable.

Now suppose that some $\rho \in X$ is not (properly) stable. Then $\lim_{t\to 0} \lambda(t) \cdot \rho$ exists for some 1-PS λ. As before, we choose a basis such that $\lambda(t) = \mathrm{diag}\,(t^{r_1}, \ldots, t^{r_n})$. Since λ is non-trivial, we have $r_a > r_{a+1}$ for some $1 \leq a < m$. But since $\lim_{t\to 0} \lambda(t) \cdot \rho$ exists, we must have that $(A_\ell)_{ij} = 0$ for all $i \geq a+1$ and $j \leq a$. Hence, ρ is a reducible representation. Conversely, one can show that if ρ is reducible, then there exists a 1-PS λ such that $\lim_{t\to 0} \lambda(t) \cdot \rho$ exists. (Should be a bit careful in defining λ, depending on what G precisely is.) We conclude that

$$X^{\mathrm{ss}} = X \qquad \text{and} \qquad X^{\mathrm{s}} = X^{\mathrm{irred.}}.$$

9 The Flag Complex

We consider the dependence of μ on λ.

Definition 9.1. Let G be a reductive algebraic group and let λ be a 1-PS. Then there is a unique algebraic subgroup $P_\lambda \subset G$ whose k-points are given by

$$P_\lambda(k) = \{p \in G : \lim_{t\to 0} \lambda(t)p\lambda(t)^{-1} \text{ exists }\}.$$

Moreover, P_λ is a *parabolic* subgroup of G, λ is a 1-PS of the *radical* of P_λ, and $\lim_{t\to 0} \lambda(t)p\lambda(t)^{-1}$ *centralizes* λ for all $p \in P_\lambda$.

Proof. Work with a faithful embedding $G \subset \mathrm{GL}_n$. Choosing a suitable basis, we may assume that λ is diagonal, i.e.

$$\lambda(t) = [t^{r_i} \cdot \delta_{ij}],$$

and we can even assume $r_1 \geq \cdots \geq r_n$. But now, for any $g = [a_{ij}] \in \mathrm{GL}_n$ we have

$$\lambda(t)g\lambda(t)^{-1} = [t^{r_i - r_j} a_{ij}],$$

which has a specialization if and only if $a_{ij} = 0$ when $r_i < r_j$, i.e. g is a block-matrix of the form

$$\left(\begin{array}{c|c|c|c} * & * & * & * \\ \hline 0 & * & * & * \\ \hline 0 & 0 & * & * \\ \hline 0 & 0 & 0 & * \end{array}\right).$$

This is known to be a parabolic subgroup of GL_n, and its radical is precisely the set of a_{ij} such that the diagonal blocks are multiples of the identity. Indeed λ is a 1-PS of this subgroup, so it is also a subgroup of the radical of P. Finally, we have

$$\lim_t \lambda(t) \cdot g = [a'_{ij}], \qquad \text{with } a'_{ij} = \begin{cases} a_{ij} & \text{if } r_i = r_j, \\ 0 & \text{otherwise} \end{cases},$$

which clearly centralizes λ. □

Definition 9.2. If T is a torus, then we define

$$\Gamma^{\mathbb{Q}}(T) = \mathrm{Hom}(\mathbb{G}_m, T) \otimes \mathbb{Q} \qquad \text{and} \qquad \Gamma^{\mathbb{R}}(T) = \mathrm{Hom}(\mathbb{G}_m, T) \otimes \mathbb{R}.$$

Note that $\mathrm{Hom}(\mathbb{G}_m, T)$ is indeed a $\mathbb{Z}$-module, as T is abelian.

In particular any 1-PS λ of T defines a point $\overline{\lambda} \in \Gamma^{\mathbb{R}}(T)$.

Proposition 9.3. *Let G be a reductive group acting on X, which is proper over k. Then for all closed points x, $\mathcal{L} \in Pic^G(X)$, and 1-PS λ, we have*

$$\mu^{\mathcal{L}}(x, \lambda) = \mu^{\mathcal{L}}(x, g\lambda g^{-1}) \qquad \textit{for all } g \in P_\lambda(k).$$

Proof. Let $y = \lim_{t \to 0} \lambda(t) \cdot x$. Then

$$\begin{aligned} \lim_{t \to 0} \left(g\lambda(t)g^{-1}\right) \cdot x &= \lim_{t \to 0} \left(g\lambda(t)g^{-1}\lambda(t)^{-1}\right) \cdot (\lambda(t) \cdot x) \\ &= g \cdot \lim_{t \to 0} \left(\lambda(t)g^{-1}\lambda(t)^{-1}\right) \cdot y \end{aligned}$$

(Finish the proof) □

Definition 9.4. Let G be a reductive group. The *rational flag complex* of G, notation $\Delta(G)$, is defined as the set of non-trivial 1-PS's λ of G, modulo the equivalence relation:

$$\lambda_1 \sim \lambda_2 \text{ if there are } n_1, n_2 > 0 \text{ and } g \in P_{\lambda_1}(k) \text{ such that } \lambda_2(t^{n_2}) = g\lambda_1(t^{n_1})g^{-1} \text{ for all } t.$$

It can be shown that this is independent of the choice of g.

Note that if $\lambda_1 \sim \lambda_2$, then $P_{\lambda_1} = P_{\lambda_2}$, so we can talk about $P(\delta)$ for $\delta \in \Delta(G)$. Furthermore, we see that $\lim_{t \to 0} \lambda_1(t) \cdot x = \lim_{t \to 0} \lambda_2(t) \cdot x$, so we obtain

$$\mu^{\mathcal{L}}(x, \lambda_1) = \frac{n_2}{n_1} \mu^{\mathcal{L}}(x, \lambda_2).$$

We are going to construct a function on $\Delta(G)$. Fix a maximal torus T of G, and let $N(T)$ be the normalizer of T. Then $N(T)/T$ is a finite group – the *Weyl group* – and acts on $\Gamma^{\mathbb{R}}(T)$ via conjugation. Fix a positive definite inner product on $\Gamma^{\mathbb{R}}(T)$ invariant under this group, which is rational on $\Gamma^{\mathbb{Q}}(T)$.

Definition 9.5. Let λ be a 1-PS of G. There exists some $g \in G$ such that $g\lambda g^{-1}$ is a 1-PS of T. This yields $\overline{g\lambda g^{-1}} \in \Gamma^{\mathbb{R}}(T)$, and we define the **norm** of λ to be

$$\|\lambda\| := \sqrt{\langle \overline{g\lambda g^{-1}}, \overline{g\lambda g^{-1}}, \rangle}.$$

Definition 9.6. For x a closed point of X, and $\delta \in \Delta(G)$, put

$$\nu^{\mathcal{L}}(x, \delta) := \mu^{\mathcal{L}}(x, \lambda)/\|\lambda\|.$$

(Define normalized weight function)

Intuitively, $\Delta(G)$ may be considered the set of rational points at ∞ on G.

10 Hesselink's / Kempf-Ness stratification of the unstable locus

Let G be a reductive group acting on a projective variety X and suppose $\mathcal{L}$ an ample linearization of this action. If $x \in X^{\mathrm{us}}$ unstable, then, by the Hilbert–Mumford criterion, there exists a 1-PS λ such that $\mu^{\mathcal{L}}(x, \lambda) < 0$. We would ideally like to see which 1-parameter subgroups(s) are most responsible for the instability of x, but since

$$\mu^{\mathcal{L}}(x, \lambda^n) = n\mu^{\mathcal{L}}(x, \lambda),$$

the quantity $\mu^{\mathcal{L}}(x, -)$ is unbounded. However if we pick a norm for 1-PSs such that $\|\lambda^n\| = n\|\lambda\|$, then

$$\frac{\mu^{\mathcal{L}}(x, \lambda)}{\|\lambda\|} = \frac{\mu^{\mathcal{L}}(x, \lambda^k)}{\|\lambda^n\|}$$

and we can instead work with this normalized Hilbert–Mumford function.

We want to choose such a norm which is invariant under conjugation by G. Now, the conjugacy classes of G are equal to the Weyl group orbits in a maximal torus T, so we can pick a norm on the set of 1-PS of a fixed maximal torus T which is invariant under the Weyl group $W = N(T)/T$ (here $N(T)$ denotes the normalizer of T). Such a norm is not hard to find: given any norm on $X_*(T) \simeq \mathbb{Z}^n$ (for some n), we can produce a norm which is W-invariant by averaging this norm over the finite group W.

Example 10.1. For $G = \mathrm{GL}_n$, we can take T the maximal torus consisting of diagonal matrices. Under the natural identification

$$\begin{aligned} \mathbb{Z}^n &\to X_*(T) \\ (m_1, \ldots, m_n) &\mapsto \mathrm{diag}(t^{m_1}, \ldots, t^{m_n}), \end{aligned}$$

the dot product on $\mathbb{Z}^n$ provides an inner product on $X_*(T)$, and hence a norm, which is W-invariant. Here, the Weyl group W is the symmetric group on n elements and its action on T corresponds to permuting the diagonal entries.

From now on, we fix a G-invariant norm $\| - \|$ on the set of conjugacy classes of 1-PSs of G.

For any $x \in X$, we define

$$M^{\mathcal{L}}(x) = \inf_{\lambda} \frac{\mu^{\mathcal{L}}(x, \lambda)}{\|\lambda\|}.$$

(This function is known as the 'stability function') (Note that inf changes to sup if we change the sign convention of μ) (Ness has a nice interpretation of $M^L(x)$ as a distance function) One can show that $M^{\mathcal{L}}(x)$ is bounded and can attain only finitely many values. This leads to the definition: a 1-PS λ for which

$$\frac{\mu^{\mathcal{L}}(x, \lambda)}{\|\lambda\|} = M^{\mathcal{L}}(x)$$

is said to be *adapted* to x. Let $\Lambda^{\mathcal{L}}(x)$ be the set of non-divisible 1-PS adapted to x. Non-divisible means that the 1-PS is not some power of another 1-PS.

Let $\langle\lambda\rangle$ be a conjugacy class of a 1-PS and let $d > 0$, then we define

$$S_{d,\langle\lambda\rangle} := \left\{x \in X : M^{\mathcal{L}}(x) = -d \text{ and } \exists g \in G : g\lambda g^{-1} \in \Lambda^{\mathcal{L}}(x)\right\}$$

which we refer to as *blades*. We can write

$$X^{\text{us}} = \bigsqcup_{d,\langle\lambda\rangle} S_{d,\langle\lambda\rangle}.$$

Once we fix a maximal torus T of G, we can pick a representative $\lambda : \mathbb{G}_m \to T$ for $\langle\lambda\rangle$. Then we write

$$S_{d,\lambda} = \{x \in X : M^{\mathcal{L}}(x) = -d \text{ and } \lambda \in \Lambda^{\mathcal{L}}(x)\}.$$

Then $S_{d,\langle\lambda\rangle} = GS_{d,\lambda}$.

Then we define the limit sets

$$Z_{d,\lambda} = \{x \in X : M^{\mathcal{L}}(x) = -d \text{ and } \lambda \in \Lambda^{\mathcal{L}}(x) \text{ and } \lambda(\mathbb{G}_m) \subset G_x\}.$$

Now, taking the limit $x \mapsto \lim_{t\to 0} \lambda(t) \cdot x$ gives the retraction

$$\pi : S_{d,\lambda} \to Z_{d,\lambda}.$$

The conjugacy classes in G correspond to the Weyl group orbit in T. Hence we can fix a representative $\lambda : \mathbb{G}_m \to T$ for each Weyl group orbit and write

$$X^{\text{us}} = \bigsqcup_{d,\lambda} GS_{d,\lambda}$$

where the union is over $d > 0$ and $\lambda \in X_*(T)/W$.

Example 10.2. Let $G = \mathrm{SL}_2$ act on $\mathbb{P}^1$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot (x : y) = (ax + by : cx + dy).$$

This induces an action of G on $X = \left(\mathbb{P}^1\right)^3$ by

$$g \cdot (p_1, p_2, p_3) = (g \cdot p_1, g \cdot p_2, g \cdot p_3).$$

The Segre embedding $\left(\mathbb{P}^1\right)^3 \to \mathbb{P}^7$ allows us to view X as a closed subvariety of projective space, and w.r.t. this embedding the action is linear. (In fact, the linearization on X is given by the exterior product $\mathcal{O}_{\mathbb{P}^1}^{\boxtimes 3}$.) We fix a maximal torus

$$T = \left\{ \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} : t \in \mathbb{C}^\times \right\} \subset G$$

and note that any 1-PS of G is conjugate to one of T. Moreover, since

$$\lambda(t) = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \qquad \text{and} \qquad \lambda^{-1}(t) = \begin{pmatrix} t^{-1} & 0 \\ 0 & t \end{pmatrix}$$

are conjugate, any 1-PS of G is conjugate to λ^r for $r \geq 1$. In particular, there is only one conjugacy class of non-divisible 1-PSs given by $\langle \lambda \rangle$. To compute the stratification, we must find the values of d for which $Z_{d,\lambda}$ is non-empty. As $Z_{d,\lambda}$ is contained in the fixed-point locus of $\lambda(\mathbb{G}_m)$, we see that $p = (p_1, p_2, p_3)$ is in $Z_{d,\lambda}$ if and only if each p_i is fixed by λ, i.e. $p_i = (1 : 0)$ or $p_i = (0 : 1)$. This gives $2^3 = 8$ points, and we list those with $\mu(p, \lambda) < 0$.

- the point $p = ((1 : 0), (1 : 0), (1 : 0))$ has $\mu(p, \lambda) = -3$,
- the points $p = ((1 : 0), (1 : 0), (0 : 1))$ and $p = ((1 : 0), (0 : 1), (1 : 0))$ and $p = ((0 : 1), (1 : 0), (1 : 0))$ have $\mu(p, \lambda) = -1$.

Using the natural norm, we have $\|\lambda\| = 1$. Then there are two unstable strata indexed by $(1, \langle \lambda \rangle)$ and $(3, \langle \lambda \rangle)$,

$$S_{1,\langle \lambda \rangle} = \{(p_1, p_2, p_3) : \text{ two of the three } p_i \text{ agree } \}$$

and

$$S_{3,\langle \lambda \rangle} = \{(p, p, p) : p \in \mathbb{P}^1\}.$$

And hence the semi-stable locus consists of mutually distinct points (p_1, p_2, p_3).

11 Variation of GIT and the GIT-fan

This section is based on this paper.

No canonical choice of $\mathcal{L}$, so people started investigating relations between different linearizations. Conclusions: finitely many quotients are possible. There is some GIT fan, chambers and walls.

Definition 11.1. Two elements $\mathcal{L}_1$ and $\mathcal{L}_2$ in $\mathrm{Pic}^G(X)$ are said to be *G-algebraically equivalent* if there exists a connected variety S, points $t_1, t_2 \in S$ and a G-linearized line bundle $\mathcal{L}$ on $S \times X$ such that $\mathcal{L}|_{\{t_1\}\times X} = \mathcal{L}_1$ and $\mathcal{L}|_{\{t_2\}\times X} = \mathcal{L}_2$. Here, G acts on $S \times X$ via its action on the second factor.

Let $\mathrm{NS}^G(X)$ denote the quotient of $\mathrm{Pic}^G(X)$ by this equivalence relation, the *G-Néron Severi group.*

Lemma 11.2. *If $\mathcal{L}_1$ and $\mathcal{L}_2$ are G-algebraically equivalent, then*

$$\mu^{\mathcal{L}_1}(x, \lambda) = \mu^{\mathcal{L}_2}(x, \lambda)$$

for all $x \in X$ and 1-PS $\lambda : \mathbb{G}_m \to G$.

Proof. Let $y = \lim_{t\to 0} \lambda(t) \cdot x$. Now $\mathbb{G}_m$ acts via λ on the fibers $\mathcal{L}_{(s,y)}$ for all $s \in S$, and the resulting map $S \to \mathbb{Z}, s \mapsto \mu^{\mathcal{L}}((s, y), \lambda)$ is clearly locally constant. From the connectedness of S then follows that $\mu^{\mathcal{L}_1}(x, \lambda) = \mu^{\mathcal{L}}((s_1, y), \lambda) = \mu^{\mathcal{L}}((s_2, y), \lambda) = \mu^{\mathcal{L}_2}(x, \lambda)$. □

We denote by $\mathrm{NS}^G(X)_{\mathbb{R}}$ the vector space $\mathrm{NS}^G(X) \otimes_{\mathbb{Z}} \mathbb{R}$, and similarly for $\mathrm{NS}^G(X)_{\mathbb{Q}}$, the subset of *rational points*. Thaddeus proved there is an exact sequence

$$0 \to \mathcal{X}^*(X)_{\mathbb{R}} \to \mathrm{NS}^G(X)_{\mathbb{R}} \to \mathrm{NS}(X)_{\mathbb{R}} \to 0.$$

The above lemma allows to define

$$\mu^\bullet(x, \lambda) : \mathrm{NS}^G(X) \to \mathbb{Z}$$

and by linearity

$$\mu^\bullet(x, \lambda) : \mathrm{NS}^G(X)_{\mathbb{R}} \to \mathbb{R}.$$

And also

$$M^\bullet(x) : \mathrm{NS}^G(X)_{\mathbb{R}} \to \mathbb{R}, \quad \ell \mapsto \sup_\lambda \frac{\mu^\ell(x, \lambda)}{\|\lambda\|}.$$

Note that $M^\bullet(x)$ is convex and positively homogeneous (i.e. $f(rx) = rf(x)$ for $r \geq 0$), since it is the suprema of a family of linear forms.

This allows us to extend semi-stable locuses for any $\ell \in \mathrm{NS}^G(X)_{\mathbb{R}}$,

$$X^{\mathrm{ss}}(\ell) = \{x \in X : M^\ell(x) \leq 0\} \quad \text{and} \quad X^{\mathrm{s}}(\ell) = \{x \in X : M^\ell(x) < 0\}. \tag{11.1}$$

Denote by $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ the convex cone generated by classes of ample G-linearized line bundles in $\mathrm{NS}^G(X)_{\mathbb{R}}$, which is an open cone (it is the preimage of the *ample cone* under the forgetful map $\mathrm{NS}^G(X)_{\mathbb{R}} \to \mathrm{NS}(X)_{\mathbb{R}}$).

We say $\ell \in \mathrm{NS}^G(X)^+_{\mathbb{R}}$ is *effective* if $X^{\mathrm{ss}}(\ell) \neq \varnothing$. The set of effective points of $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ is denoted $C^G(X)$, the *G-ample cone*, which can be shown to be convex.

Two points $\ell, \ell' \in \mathrm{NS}^G(X)^+_{\mathbb{R}}$ are called *GIT-equivalent* if $X^{\mathrm{ss}}(\ell) = X^{\mathrm{ss}}(\ell')$. The *GIT-class* of any ℓ is the set of all ℓ' that are GIT-equivalent. We are interested in the map

$$(\ell \in C^G(X)) \mapsto X^{\mathrm{ss}}(\ell).$$

Remark 11.3. For rational $\ell \in \mathrm{NS}^G(X)_{\mathbb{Q}}$ there exists a G-linearized line bundle $\mathcal{L}$ on X such that $X^{\mathrm{ss}}(\ell) = X^{\mathrm{ss}}(\mathcal{L})$ and $X^{\mathrm{s}}(\ell) = X^{\mathrm{s}}(\mathcal{L})$. Indeed, from (11.1) it is easy to see that $\mathcal{L} = n\ell \in \mathrm{NS}^G(X)$ for large enough $n \in \mathbb{Z}$ suffices.

Proposition 11.4. *For all $\ell_0 \in \mathrm{NS}^G(X)^+_{\mathbb{R}}$ there exists a neighborhood V of ℓ_0 such that $X^s(\ell_0) \subset X^s(\ell) \subset X^{ss}(\ell) \subset X^{ss}(\ell_0)$ for all rational $\ell \in V$.*

Proof. By Theorem **??**, there are finitely many open subsets $X^{\mathrm{s}}_1, \dots, X^{\mathrm{s}}_n$ such that $X^{\mathrm{s}}(l)$ is one of them for all $\ell \in \mathrm{NS}^G(X)^+_{\mathbb{Q}}$. We order these sets such that

$$X^{\mathrm{s}}(\ell_0) \not\subset X^{\mathrm{s}}_i \text{ for } i = 1, \dots, p \qquad \text{and} \qquad X^{\mathrm{s}}(\ell_0) \subset X^{\mathrm{s}}_i \text{ for } i = p+1, \dots, n.$$

Fix some points $x_1, \dots, x_p \in X^{\mathrm{s}}(\ell_0)$ with $x_i \notin X^{\mathrm{s}}_i$ for all $i = 1, \dots, p$. By continuity of the functions $M^\bullet(x)$ there exist neighborhoods $V_{x_1}, \dots, V_{x_p}$ of ℓ_0 in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ such that $x_i \in X^{\mathrm{s}}(\ell)$ for all (rational) $\ell \in V_{x_i}$. If we let $V = \bigcap_{i=1}^p V_{x_i}$, then for all rational $\ell \in V$, the $X^{\mathrm{s}}(\ell)$ is different from $X^{\mathrm{s}}_1, \dots, X^{\mathrm{s}}_p$ as it contains $x_1, \dots, x_p$, so $X^{\mathrm{s}}(\ell)$ must be one of $X^{\mathrm{s}}_{p+1}, \dots, X^{\mathrm{s}}_n$, and thus contains $X^{\mathrm{s}}(\ell_0)$.

For the last inclusion: let $(B_n)_{n\in\mathbb{N}}$ be a sequence of balls centered at ℓ_0 and contained in V, whose radii converge to zero. Let $X^{\mathrm{ss}}_1, \dots, X^{\mathrm{ss}}_k$ be the open subsets of X such that $X^{\mathrm{ss}}(\ell)$ is one of them for all $\ell \in \mathrm{NS}^G(X)^+_{\mathbb{Q}}$. Note that

$$I_n = \{i : 1 \le i \le k \text{ and } X^{\mathrm{ss}}_i = X^{\mathrm{ss}}(\ell) \text{ for some rational } \ell \in B_n \text{ with } \ell \neq \ell_0\}$$

is a decreasing sequence of finite sets, so it must be stationary for some $n \ge N$. Now B_N is the neighborhood of ℓ_0 we are looking for, since for any rational $\ell \in B_N$ and $x \in X^{\mathrm{ss}}(\ell)$, we can construct a sequence $\ell_n \in B_n$ with $x \in X^{\mathrm{ss}}(\ell_n) = X^{\mathrm{ss}}(\ell)$. Since $\ell_n \to \ell_0$ and $M^\bullet(x)$ is continuous, we have $M^{\ell_0}(x) \le 0$, so $x \in X^{\mathrm{ss}}(\ell_0)$. Therefore, $X^{\mathrm{ss}}(\ell) \subset X^{\mathrm{ss}}(\ell_0)$. □

Definition 11.5. For any $x \in X$, we define the *stability set* of x to be

$$\Omega(x) = \{\ell \in \mathrm{NS}^G(X)^+_{\mathbb{R}} : x \in X^{\mathrm{ss}}(\ell)\} = \{\ell \in \mathrm{NS}^G(X)^+_{\mathbb{R}} : M^\ell(x) \le 0\}.$$

Note that since $M^\bullet(x)$ is a convex and positively homogeneous function, the set $\Omega(x)$ is a convex cone and closed in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$.

Proposition 11.6. *Let $x \in X$. The span of the stability set $\Omega(x)$ is a rational vector subspace of $\mathrm{NS}^G(X)_{\mathbb{R}}$. In particular, $\Omega(x)$ the closure of its rational points $\Omega(x) \cap \mathrm{NS}^G(X)_{\mathbb{Q}}$.*

Now note that the number of stability sets is finite. Namely, every stability set is a union of GIT-classes, so by Theorem **??** there are only finitely many sets of the form $\Omega(x) \cap \mathrm{NS}^G(X)_{\mathbb{Q}}$ for $x \in X$. The above proposition shows that $\Omega(x)$ is the closure of such a set, hence finitely many.

In particular, we find that

$$C^G(X) = \bigcup_{x\in X} \Omega(x)$$

is a finite union of closed sets, and thus closed in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$.

Lemma 11.7. *Let $x \in X$ and $z \in \overline{G \cdot x} - G \cdot x$, and assume there exists a rational point $\ell_0 \in C^G(X)$ such that $G \cdot z$ is closed in $X^{ss}(\ell_0)$. Then, there exists a 1-PS λ such that*

(i) $\lim_{t \to 0} \lambda(t) \cdot x \in G \cdot z$,

(ii) $\Omega(x) \subset \{\ell \in \mathrm{NS}^G(X)_{\mathbb{R}}$ *such that* $\mu^\ell(x, \lambda) \leq 0\}$,

(iii) $\Omega(z) = \{\ell \in \mathrm{NS}^G(X)_{\mathbb{R}}$ *such that* $\mu^\ell(x, \lambda) = 0\} \cap \Omega(x)$.

Proof. The Hilbert–Mumford theorem applied to $X^{\mathrm{ss}}(\ell_0)$ gives a 1-PS λ such that (i), and (ii) is obvious. Denote the limit by $z' = \lim_{t \to 0} \lambda(t) \cdot x$. Since λ fixes z', if $z \in X^{\mathrm{ss}}(\ell)$ then both $\mu^\ell(z', \lambda)$ and $\mu^\ell(z', -\lambda) = -\mu^\ell(z', \lambda)$ must be non-positive and thus zero, which shows $\Omega(z) = \Omega(z') \subset \{\ell$ such that $\mu^\ell(z', \lambda) = \mu^\ell(x, \lambda) = 0\}$. Now let ℓ be a rational point in $\Omega(z)$. Since $X^{\mathrm{ss}}(\ell)$ is open and $z \in \overline{G \cdot x}$, we have $g \cdot x \in X^{\mathrm{ss}}(\ell)$ for some $g \in G$, and thus $\ell \in \Omega(g \cdot x) = \Omega(x)$. But since stability sets are the closure of their rational points **??**, we have $\Omega(z) \subset \Omega(x)$. This shows the first inclusion of (iii). Conversely, let ℓ be a rational point in $\Omega(x)$ such that $\mu^\ell(x, \lambda) = 0$. Then **??** shows that $z \in X^{\mathrm{ss}}(\ell)$, that is $\ell \in \Omega(z)$. By the density of rational points in $\{\ell$ such that $\mu^\ell(x, \lambda) = 0\} \cap \Omega(x)$, we conclude the other inclusion, and thus (iii). □

Definition 11.8. A *polyhedral cone* in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ is a subset of $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ defined by a finite number of linear inequalities. Such a cone is *rational* if the inequalities can be chosen to be rational. If C is such a cone, and f a linear form non-negative on C, then the set of points c with $f(c) = 0$ is called a *face* of C. (Note that C is a face of C itself.)

Proposition 11.9. *Let $x \in X$.*

(i) There exists $y \in \overline{G \cdot x}$ such that: ℓ belongs to the relative interior of $\Omega(y)$ iff $G \cdot y$ is closed in $X^{ss}(\ell)$. Moreover $\Omega(x) = \Omega(y)$.

(ii) $\Omega(x)$ is a convex rational polyhedral cone in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$.

(iii) The faces of $\Omega(x)$ are exactly the sets $\Omega(y)$ with $y \in \overline{G \cdot x}$.

Proof. (Write proof) □

Definition 11.10. A point x is said to be *pivotal for* ℓ if $x \in X^{\mathrm{ss}}(\ell)$ and $G \cdot x$ is closed in $X^{\mathrm{ss}}(\ell)$. A point x is said to be *pivotal for* Ω if $\Omega = \Omega(x)$ and x is pivotal for some (and thus any) point in the relative interior of Ω.

Lemma 11.11. *Let $\ell_1, \ell_2 \in C^G(X)$ such that $X^{ss}(\ell_1) \subset X^{ss}(\ell_2)$, and let x be a pivotal point for ℓ_2. Then there exists a pivotal point y for ℓ_1 such that $x \in \overline{G \cdot y}$. Moreover, $X^s(\ell_2) \subset X^s(\ell_1)$.*

Proof. Consider the commutative diagram:

$$\begin{tikzcd} X^{\mathrm{ss}}(\ell_1) \arrow[r, hook] \arrow[d, "\pi_1"] & X^{\mathrm{ss}}(\ell) \arrow[d, "\pi_2"] \\ X^{\mathrm{ss}}(\ell_1) /\!\!/ G \arrow[r, "\phi"] & X^{\mathrm{ss}}(\ell_2) /\!\!/ G \end{tikzcd}$$

The image of ϕ equals $\pi_2(X^{\mathrm{ss}}(\ell_1))$, and since π_2 is surjective and $X^{\mathrm{ss}}(\ell)$ is dense (why?) in $X^{\mathrm{ss}}(\ell_2)$, ϕ is dominant. As $X^{\mathrm{ss}}(\ell_1) /\!\!/ G$ is complete, ϕ is even surjective. Therefore, there exists $y \in X^{\mathrm{ss}}(\ell_1)$ with $\phi(\pi_1(y)) = \pi_2(x)$. Since $y \in \pi_2^{-1}(\pi_2(x))$ and $G \cdot x$ is the unique closed orbit in $\pi_2^{-1}(\pi_2(x))$, we have $x \in \overline{G \cdot y}$. Moreover, $x \in \overline{G \cdot y'}$ for all $y' \in X^{\mathrm{ss}}(\ell_1) \cap \overline{G \cdot y}$. Therefore we can find y such that $G \cdot y$ is closed in $X^{\mathrm{ss}}(\ell_1)$ and $x \in \overline{G \cdot y}$. This proves the first assertion of the lemma. Let x be a stable point for ℓ_2 and let$y$ be as above. Since G_x is finite, $G \cdot x = G \cdot y$ and x is semistable forℓ_1. Moreover, $G \cdot x$ is closed in $X^{\mathrm{ss}}(\ell_2)$ and hence in $X^{\mathrm{ss}}(\ell_1)$. Thus, x is stable for ℓ_1. □

Lemma 11.12. *Let F be a GIT-class. Then*

$$F = \bigcap_{x \text{ is pivotal for } F} \mathrm{RelInt}(\Omega(x)).$$

Proof. (Write proof) □

Corollary 11.13. *Any GIT-class is the relative interior of a rational polyhedral cone in* $\mathrm{NS}^G(X)^+_{\mathbb{R}}$.

Proposition 11.14. *Let F and F' be two GIT-classes. The following are equivalent:*

(i) F' intersects the closure of F in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$

(ii) F' is contained in the closure of F in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$

(iii) $X^{ss}(F) \subset X^{ss}(F')$

Proof. Note that it suffices to prove

$$\ell \in \overline{F} \iff X^{\mathrm{ss}}(F) \subset X^{\mathrm{ss}}(\ell),$$

where $\overline{F}$ denotes the closure of F in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$. For any $\ell \in \overline{F}$ we have $X^{\mathrm{ss}}(F) \subset X^{\mathrm{ss}}(\ell)$ by continuity of the functions $M^\bullet(x)$. Conversely, take ℓ such that $X^{\mathrm{ss}}(F) \subset X^{\mathrm{ss}}(\ell)$. We use Lemma **??** to characterize F. Take any $\ell_0 \in F$ and pivotal point x for F. Then $\ell_0 \in \mathrm{RelInt}(\Omega(x))$, and $\overline{F} \subset \Omega(x)$. Hence, the segment $[\ell_0, \ell)$ is contained in $\mathrm{RelInt}(\Omega(x))$. Hence $[\ell_0, \ell) \subset F$ and thus $\ell \in \overline{F}$. □

Proposition 11.15. *The relative interior of a face of a GIT-class is a GIT-class.*

Proof. (Write proof) □

12 Global Geometry of the GIT-classes

Definition 12.1. A *wall* is the stability set of codimension one in $C^G(X)$. A *chamber* is a GIT-class of codimension zero in $C^G(X)$.

Remark 12.2. Note that by continuity of the functions $M^\bullet(x)$, the GIT-classes such that $X^{\mathrm{ss}} = X^{\mathrm{s}}$ are chambers (but not conversely).

Proposition 12.3. *The chambers are the connected components of the complement in $C^G(X)$ of the union of the walls. The GIT-classes are the relative interiors of the faces of the chambers.*

Definition 12.4. A *fan* Δ in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ is a finite set of rational convex polyhedral cones in $\mathrm{NS}^G(X)^+_{\mathbb{R}}$ such that (i) each face of a cone in Δ is also a cone in Δ, (ii) the intersection of two cones in Δ is a face of each of them.

We can summarize the above in the following theorem.

Theorem 12.5. *Let G be a reductive algebraic group acting on a normal projective variety X. Then*

(i) For all $\ell_0 \in C^G(X)$,

$$C(\ell_0) = \{\ell \in C^G(X) \text{ such that } X^{ss}(\ell_0) \subset X^s(\ell)\}$$

is a closed convex rational polyhedral cone in $C^G(X)$.

(ii) The cones $C(\ell)$ form a fan, covering $C^G(X)$.

(iii) The GIT-classes are the relative interior of these cones.

This fan is called the GIT-fan *for the action of G on X.*

Remark 12.6. Along a wall crossing, the unstable and stable points do not change by continuity of $M^\bullet(x)$. However, the set of semi-stable points might change.